Deep Learning Systems Analysis Report

Traffic Sign Recognition with a CNN, Capstone Project 4 (Deep Learning Systems)

Report Overview

This report presents an image-classification system that identifies German traffic signs from photographs using a Convolutional Neural Network (CNN) built in PyTorch. The dataset is the German Traffic Sign Recognition Benchmark (GTSRB), 43 classes of real road-sign photographs. A baseline CNN was trained first, then a second version of the same model was trained with one change, the addition of data augmentation, so the two models could be compared. The full implementation is in the accompanying notebook (`deep_learning.ipynb`).

Dataset and Task Description

GTSRB is a benchmark dataset of real photographs of German road signs across 43 classes, collected from actual road traffic to support traffic sign recognition research (Stallkamp, Schlipsing, Salmen, & Igel, 2012). The data was obtained from the `tanganke/gtsrb` Hugging Face dataset mirror, which packages the GTSRB images and labels into the same published train/test split structure (26,640 training images, 12,630 test images). The training split was further divided 85/15 into 22,644 training images and 3,996 validation images used to monitor training progress; the 12,630-image test split was never touched until final evaluation. Every image was resized to a fixed 32x32 resolution and scaled to the [0, 1] pixel range so that images of varying original size and aspect ratio could be batched into fixed-size tensors. This is a supervised, 43-class image classification task, well suited to a CNN because convolution exploits the local, spatially repeating structure of a photographed sign (edges, color blocks, symbols) in a way that a fully connected network applied directly to raw pixels would not (LeCun, Bottou, Bengio, & Haffner, 1998).

Model Architecture and Design Decisions

The baseline model (`TrafficSignCNN`) has three convolution-ReLU-max pool blocks that increase channel depth while halving spatial resolution at each step (3→32→64→128 channels, 32x32→16x16→8x8→4x4), the standard convolutional feature-extraction pattern for small, fixed-size images (LeCun et al., 1998). ReLU activations were used throughout rather than sigmoid or tanh, since ReLU is computationally cheap and does not saturate for positive inputs, which keeps gradients from vanishing as they propagate back through

multiple layers. The resulting 128x4x4 feature map is flattened and passed through a 256-unit fully connected layer with a ReLU activation, then a final linear layer produces 43 raw class scores. No softmax is applied inside the model itself, because the training loss function, cross-entropy, applies it internally; cross-entropy was chosen because it is the standard loss for multi-class, mutually exclusive classification. The optimizer is Adam with a learning rate of 1e-3, chosen because it adapts its per-parameter step size using running estimates of gradient moments, which converges reliably on this kind of task without requiring extensive manual learning-rate tuning (Kingma & Ba, 2015). Both the baseline and the experimental model in this report use the identical architecture, optimizer, learning rate, loss function, batch size, number of epochs (10), and weight-initialization seed, so the two runs are contrasted on a single, controlled dimension.

Experimental Comparison

The single change introduced in the second run is data augmentation applied to the training images only: random rotation (±15°), random translation and scaling (±10%), and random brightness/contrast jitter. Validation and test images were never augmented in either run, and the train/validation split indices were identical across both runs, so any difference in results can only be attributed to what the augmented model saw during training. This change was chosen over an architecture or hyperparameter change because road signs are photographed in practice from varying distances, angles, and lighting; augmentation is a standard way to expose a model to that kind of variation synthetically during training, which can reduce overfitting to the exact pixel arrangements seen in the training set and improve generalization to new, unseen photographs (Shorten & Khoshgoftaar, 2019). It is a meaningful comparison specifically because it isolates a training-data strategy rather than a model-capacity change, testing whether synthetic viewpoint/lighting variation helps more than simply memorizing the training images more precisely.

Results and Interpretation

Table 1 reports accuracy and macro-F1 (an unweighted average across all 43 classes, which keeps common classes such as speed limits from drowning out rarer sign types) on the held-out test set for both models.

Model	Final Validation Accuracy	Test Accuracy	Test Macro-F1
Baseline	98.35%	81.96%	0.751

Baseline + Augmentation	97.70%	81.11%	0.752

The two models are essentially tied on the test set: the baseline has a marginally higher raw accuracy, the augmented model a marginally higher macro-F1, and both differences (0.85 percentage points of accuracy, 0.001 of macro-F1) are well within the run-to-run noise expected from a single training run. Augmentation also made training visibly slower to converge (Figure 1): the augmented model's training and validation loss curves lag behind the baseline's at every epoch, which is expected since augmented images are a harder training signal to fit, but after 10 epochs it had not caught up enough to produce a clearly better generalizing model in this setup.

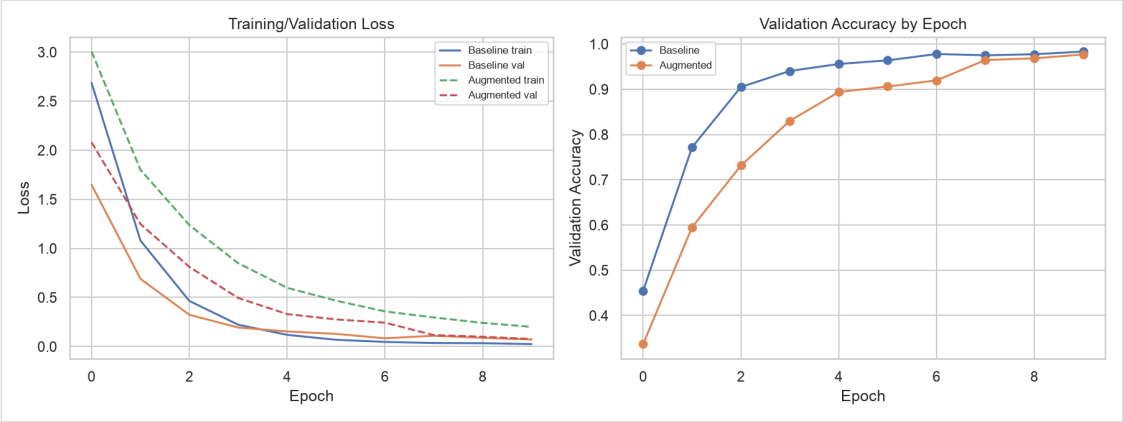


Figure 1. Training/validation loss and validation accuracy by epoch for both models. The augmented model's slower convergence is visible in both panels.

The most informative result in this experiment was not the augmentation comparison itself but the gap between validation and test performance that both models show: validation accuracy reached 98.35% (baseline) and 97.70% (augmented), roughly 16 percentage points higher than either model's accuracy on the true held-out test set. This is a concrete instance of overfitting to the validation split rather than to the training loss curve directly, and it has a specific, identifiable cause here: GTSRB's training images are captured as short video tracks of each physical sign instance, so a single sign produces many near-duplicate consecutive frames (Stallkamp et al., 2012). A random 85/15 split places frames from the same track on both sides, so the validation set is not an independent sample of sign instances the way the official test split is, and validation accuracy overstates how well either model generalizes to signs it has truly never seen.

Figure 2 shows one concrete behavior example drawn directly from the notebook's outputs: a photograph whose true label is a 20 km/h speed limit sign. The baseline model misclassified it as a 30 km/h speed limit sign; the augmented model classified it correctly.

This single case matches a broader pattern visible in both models' confusion matrices, where most of the test-set errors fall between adjacent speed-limit classes (for example 20/30 km/h or 80/100 km/h), which look nearly identical at a resized 32x32 resolution, since they differ only in one or two digits occupying a small fraction of the sign.



Figure 2. A test image the baseline model misclassified (true: 20 km/h speed limit, predicted: 30 km/h) that the augmented model classified correctly.

Limitations and Risks

Validation/test distribution mismatch: as discussed above, the 16-point validation-to-test accuracy gap shows that this notebook's validation split, built by randomly dividing the training images, is not an unbiased estimate of true generalization performance, because of the near-duplicate track structure in the GTSRB training data. Model-selection decisions made using validation accuracy alone (for example, early stopping or hyperparameter choice) would be more trustworthy if the validation split instead held out entire sign tracks rather than individual frames.

Resolution loss: every image was resized to 32x32 pixels regardless of its original resolution, which discards fine detail that could matter for visually similar sign pairs such as adjacent speed limits; the misclassification example in Figure 2 is consistent with this limitation.

No ROI cropping: unlike the original benchmark evaluation protocol, which can use the sign's bounding box, images here were used as full uncropped photographs, so some surrounding background and other scene content is present in every input and was not controlled for.

Single run per configuration: both models were trained once each rather than averaged over multiple seeds, so the small (under 1 percentage point) difference between the baseline and augmented test results should not be read as a reliable effect; a repeated-seed comparison would be needed to say with confidence whether augmentation helps, hurts, or is neutral for this architecture and training budget.

Ethical and Responsible Use

The concrete error found in this notebook, confusing a 20 km/h speed limit sign for a 30 km/h sign, is not a cosmetic mistake if this kind of classifier were ever used as an input to a real driver-assistance or autonomous-driving system: under-reading a speed limit could encourage unsafe speed in a zone with a lower limit for a real safety reason (a school, a residential street), while over-reading one could trigger unnecessary alerts or braking. Because the test-set confusion pattern shows this error concentrated specifically among visually similar, adjacent-value signs rather than spread randomly across all 43 classes, any deployment of a model like this in an ADAS context would need per-class error analysis, not just an aggregate accuracy number, before it could be trusted for any safety-relevant decision, and should never be the sole input used to override a human driver's own judgment of a sign's value.

Future Improvements

With more time and compute, a track-aware train/validation split (holding out entire sign-instance tracks rather than individual frames) would give a validation metric that actually predicts test performance, closing the gap identified above. Training for more epochs with a learning-rate schedule, and averaging results over multiple random seeds, would also clarify whether the augmentation change in this report is truly neutral or whether 10 epochs was simply not enough training budget for the augmented model to catch up. A deeper

architecture, or cropping to the sign's bounding box before resizing, could also be tried as the next controlled comparison, building on the single-variable methodology used here.

References

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